Game Theory

Dylan Losey


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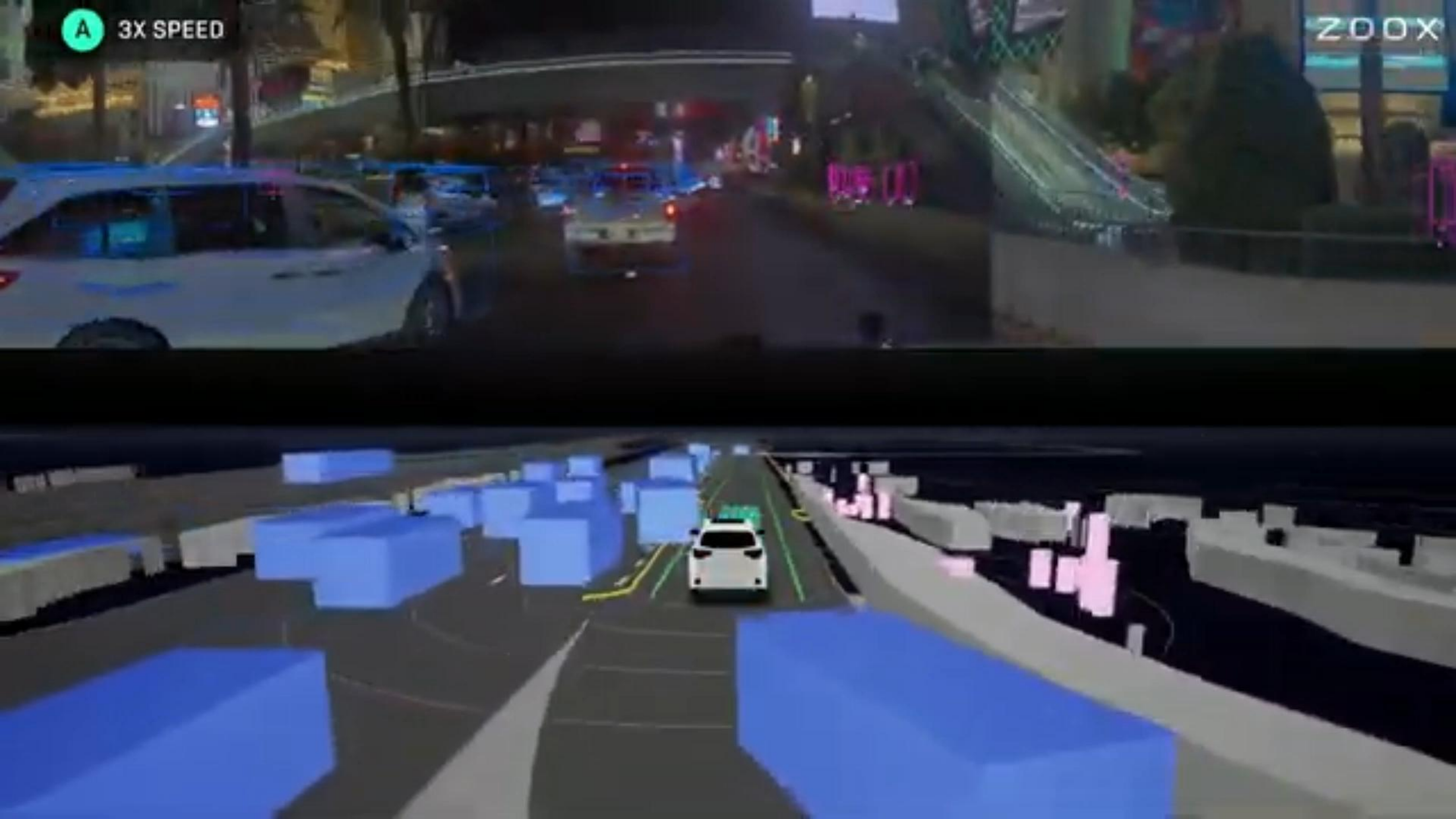
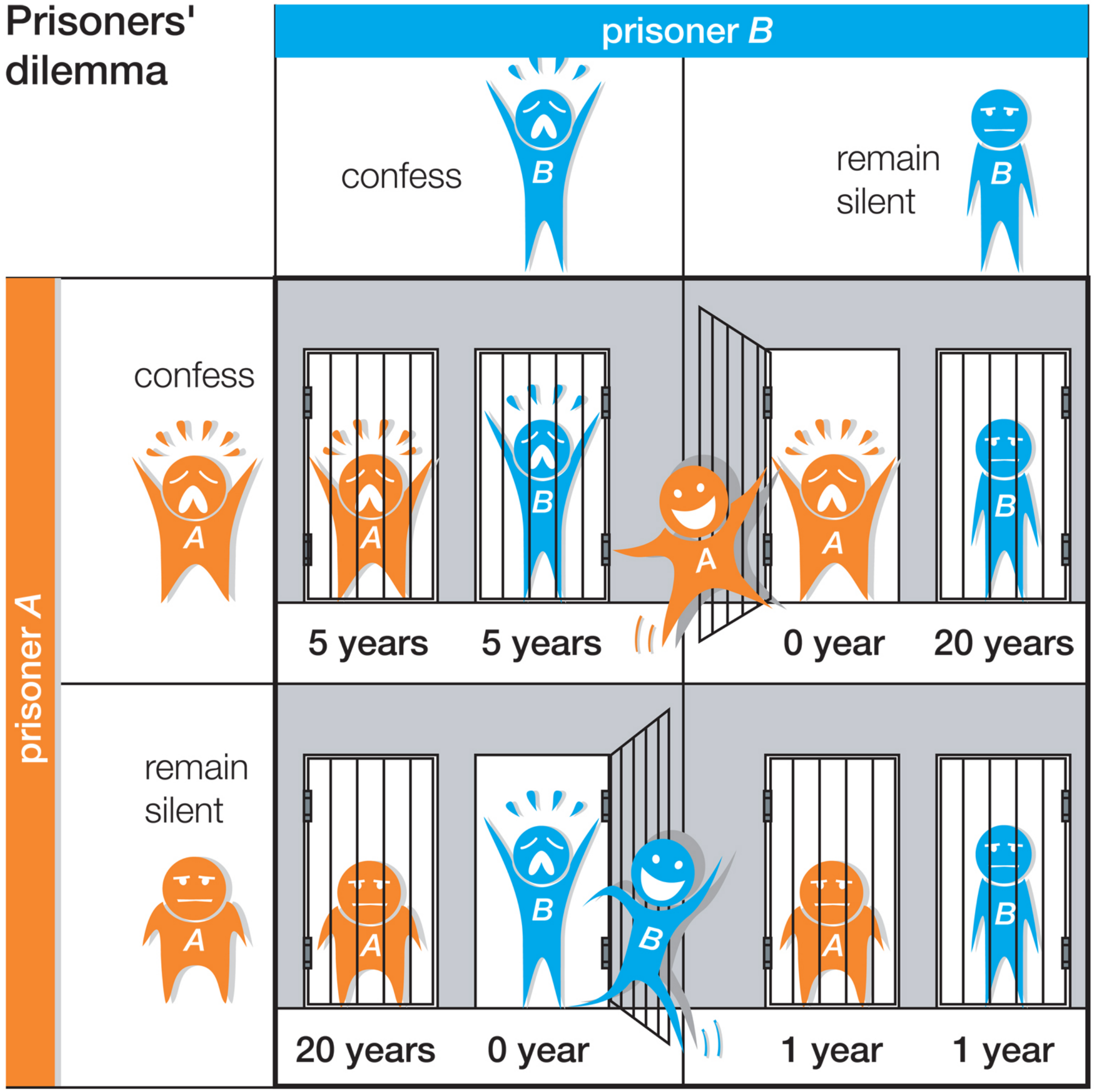

A
3X SPEED

This Lecture

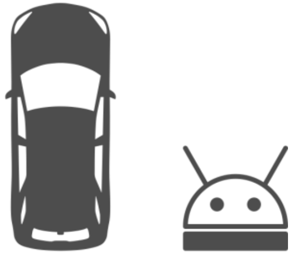
- How do we write human-robot interaction as a two-player game?
- Using game theory to influence humans

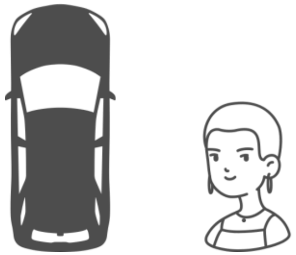


Prisoners' dilemma
prisoner B
confess
remain silent
prisoner A
confess
remain silent
5 years
5 years
0 year
20 years
20 years
0 year
1 year
1 year
A
B


How can we write HRI as
a two-player game?

Initial state s^0





Multi-Agent Games

Consider a human-robot system with **dynamics**:

$$s^{t+1} = f\left(s^t, a_R^t, a_H^t\right)$$

system state

human action

robot action

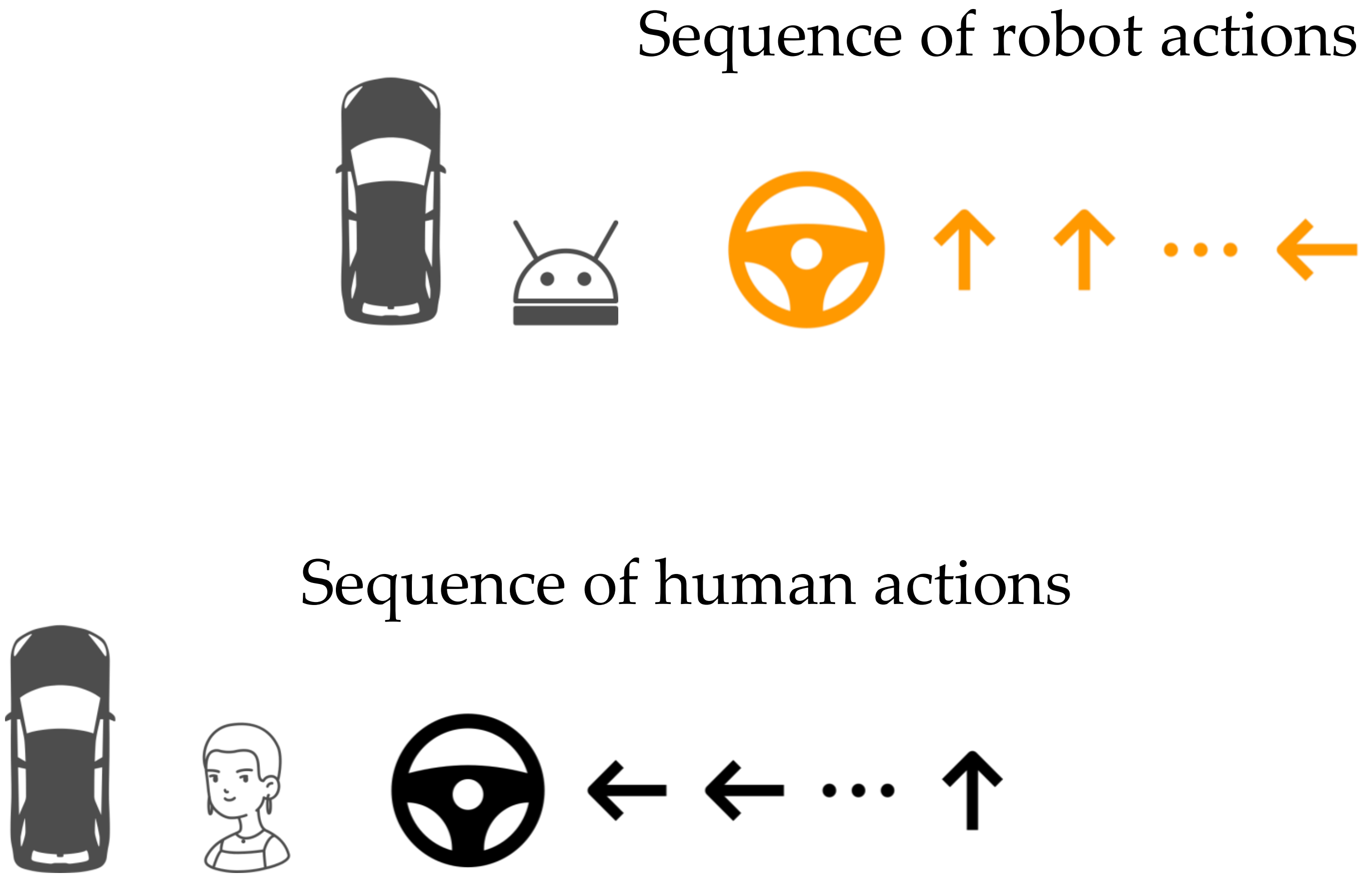
Multi-Agent Games

Consider a human-robot system with **dynamics**:

$$s^{t+1} = f\left(s^t, a_R^t, a_H^t\right)$$

Let's assume the dynamics are **deterministic**. Over T timesteps:

- $\boldsymbol{a_R} = \left(a_R^0, a_R^1, \dots, a_R^T\right)$ is the sequence of robot actions
- $\boldsymbol{a_H} = \left(a_H^0, a_H^1, \dots, a_H^T\right)$ is the sequence of human actions


Sequence of robot actions
Sequence of human actions

Multi-Agent Games

The human and robot have **separate** reward functions

$$R_H(s^0, \boldsymbol{a_R}, \boldsymbol{a_H}) = \sum_{t=0}^{T} r_H(s^t)$$

$$R_R(s^0, \boldsymbol{a_R}, \boldsymbol{a_H}) = \sum_{t=0}^{T} r_R(s^t)$$

Multi-Agent Games

The human and robot have **separate** reward functions

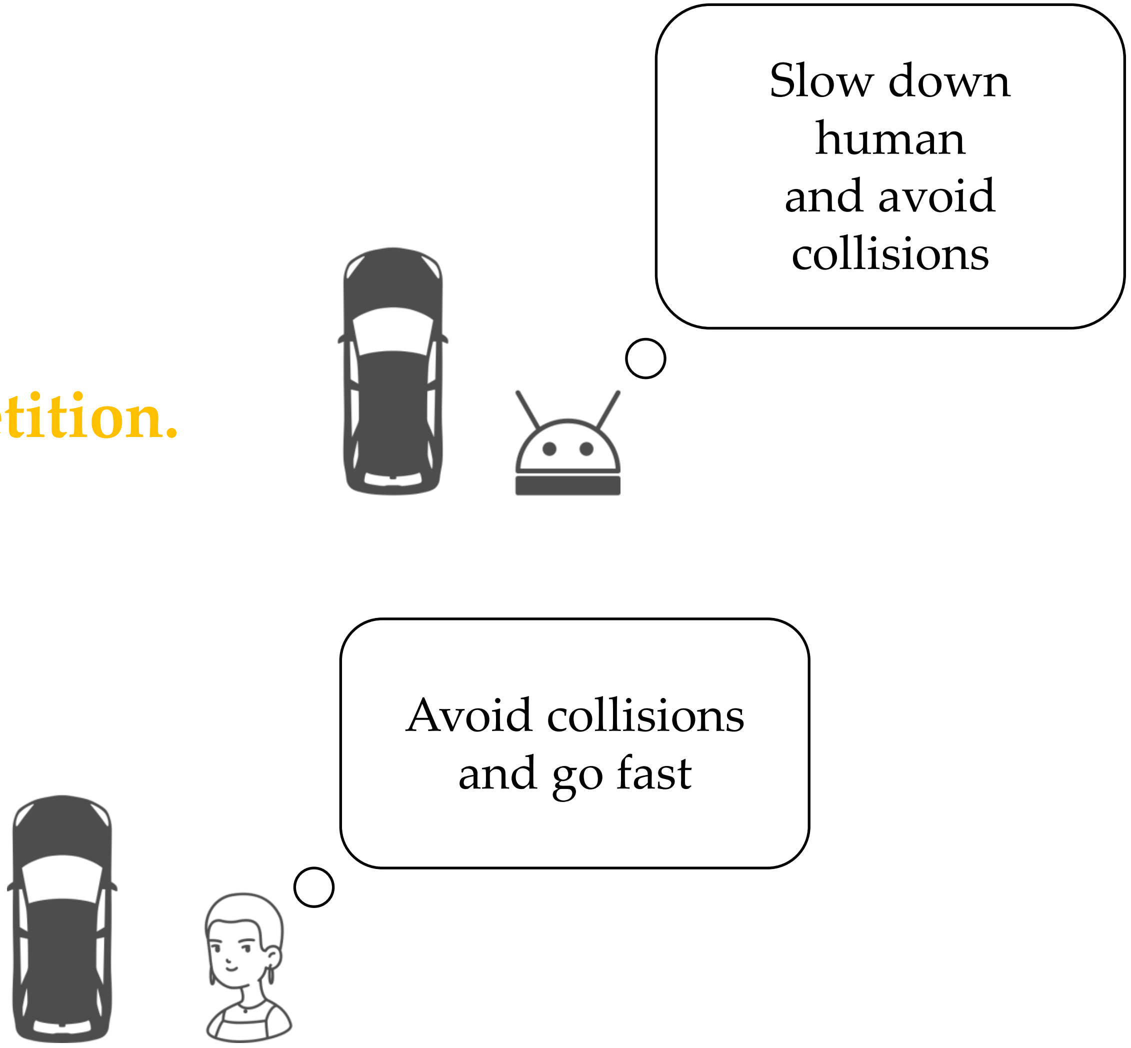
1. **Collaboration**. Both agents get same reward at every timestep.

$$R_H(s^0, \boldsymbol{a_R}, \boldsymbol{a_H}) = R_R(s^0, \boldsymbol{a_R}, \boldsymbol{a_H})$$

2. **Competition**. Agents get different rewards.

$$R_H(s^0, \boldsymbol{a_R}, \boldsymbol{a_H}) \neq R_R(s^0, \boldsymbol{a_R}, \boldsymbol{a_H})$$

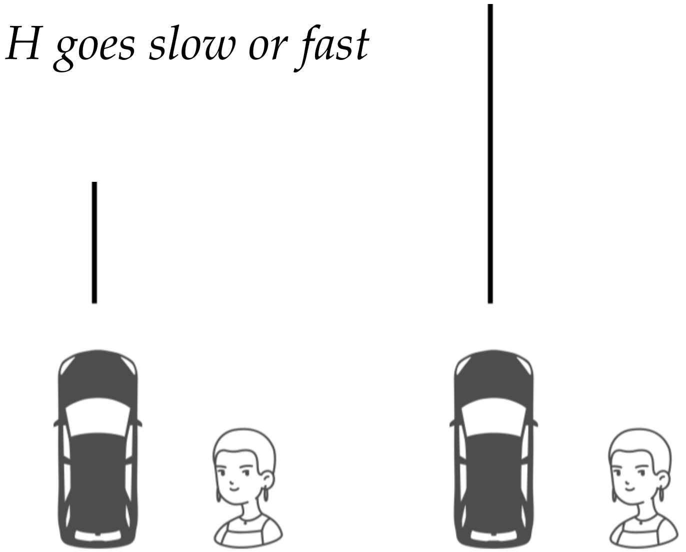
Often we have a mix of **collaboration** and **competition.**

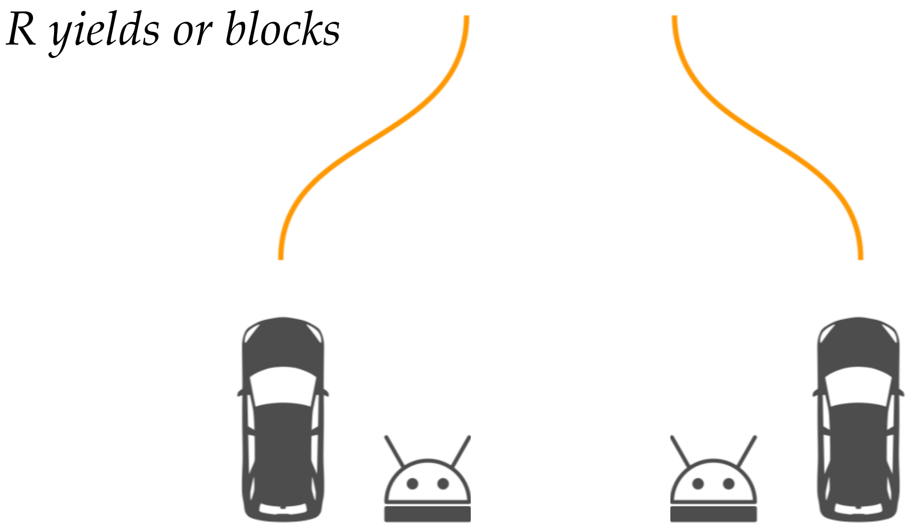



Robots should reason over
how humans will react.

Nash Equilibrium

For now let's simplify the problem so that each agent takes a single action.





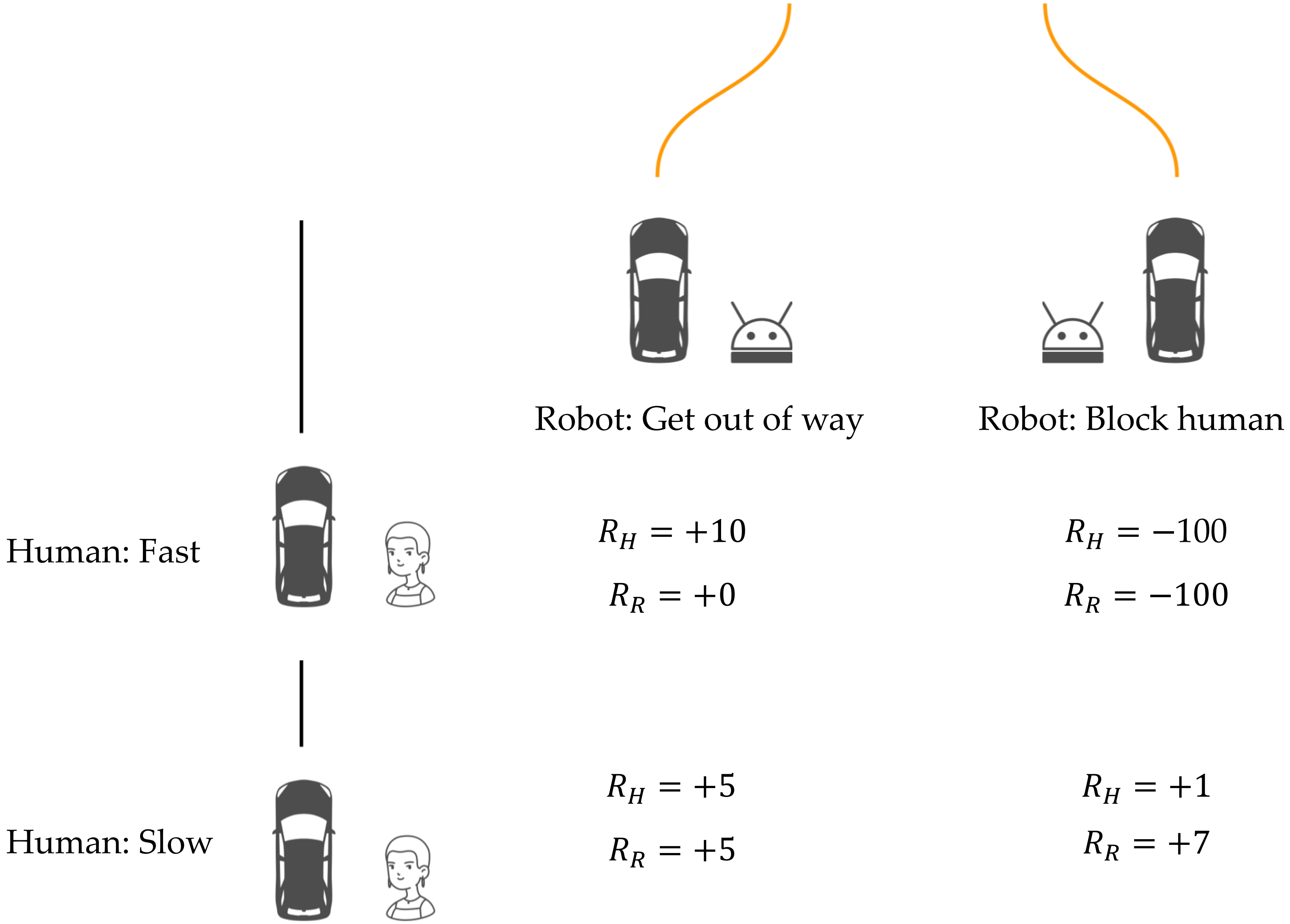
Nash Equilibrium

Nash Equilibrium.

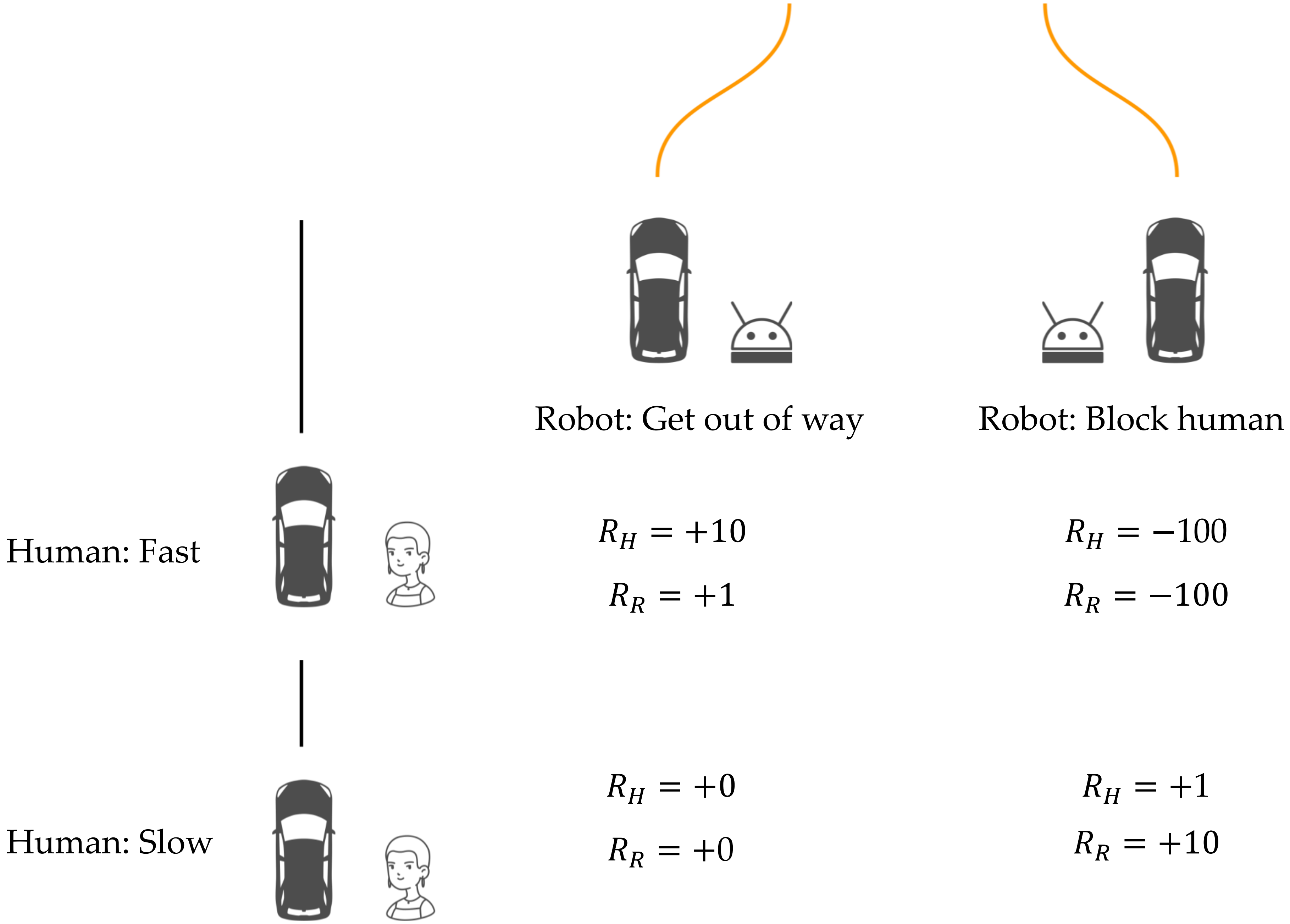
Pair of human and robot policies where, if either agent changes their policy, they will receive less expected reward.

Notes.

- For a given problem there can be more than one Nash Equilibrium
- When we are uncertain about the other agent's policy, we play the policy that works best in expectation



Robot: Get out of way
Robot: Block human
Human: Fast
$R_H = +10$
$R_R = +0$
$R_H = -100$
$R_R = -100$
Human: Slow
$R_H = +5$
$R_R = +5$
$R_H = +1$
$R_R = +7$


Robot: Get out of way
Robot: Block human
Human: Fast
$R_H = +10$
$R_R = +1$
$R_H = -100$
$R_R = -100$
Human: Slow
$R_H = +0$
$R_R = +0$
$R_H = +1$
$R_R = +10$

How can robots plan while accounting for human responses?

Robot optimizes for

$R_R(s^0, \boldsymbol{a_R}, \boldsymbol{a_H})$

Sequence of robot actions

Human optimizes for

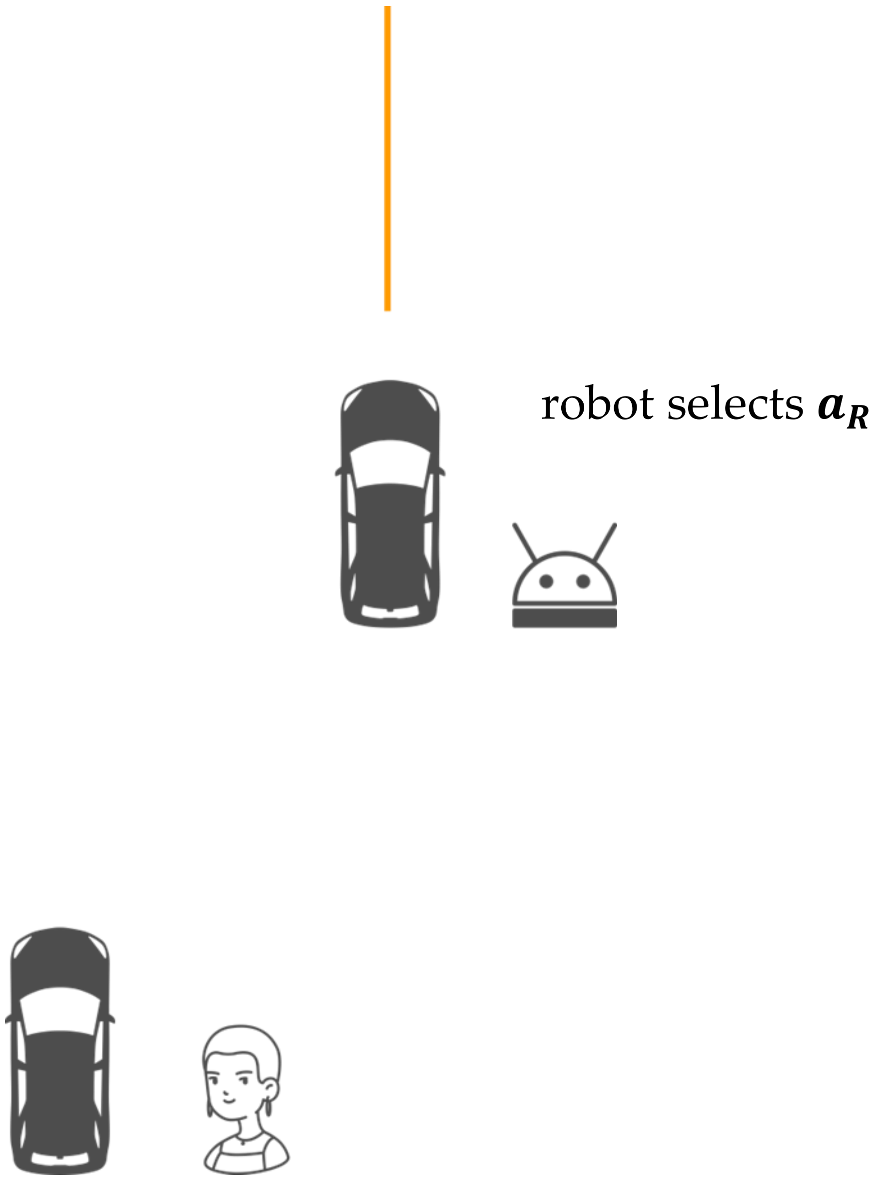
$R_H(s^0, \boldsymbol{a_R}, \boldsymbol{a_H})$

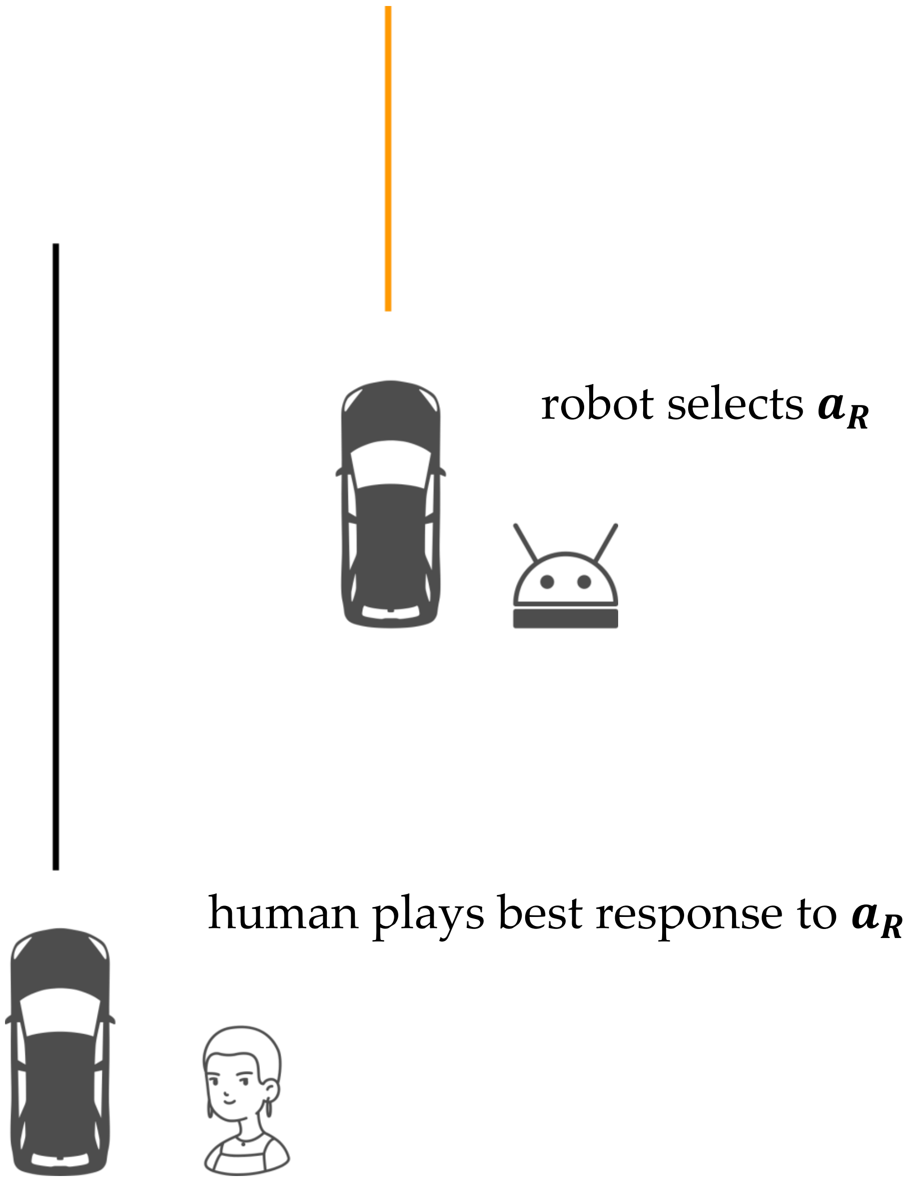
Sequence of human actions

Stackelberg Games

In a **Stackelberg game** one agent gets to play first.

The other agent observes their actions, and then choses a best response.


robot selects a_R


robot selects $\boldsymbol{a_R}$
human plays best response to $\boldsymbol{a_R}$

Stackelberg Games

In **Stackelberg game** robot plays first and human chooses best response.

$$a_H^* = \text{argmax}_{\boldsymbol{a_H}} R_H(s^0, \boldsymbol{a_R}, \boldsymbol{a_H})$$

Given robot selection $\boldsymbol{a_R}$
human finds optimal actions $\boldsymbol{a_H^*}$
Note that $\boldsymbol{a_H^*}$ *depends on the robot's choice* $\boldsymbol{a_R}$

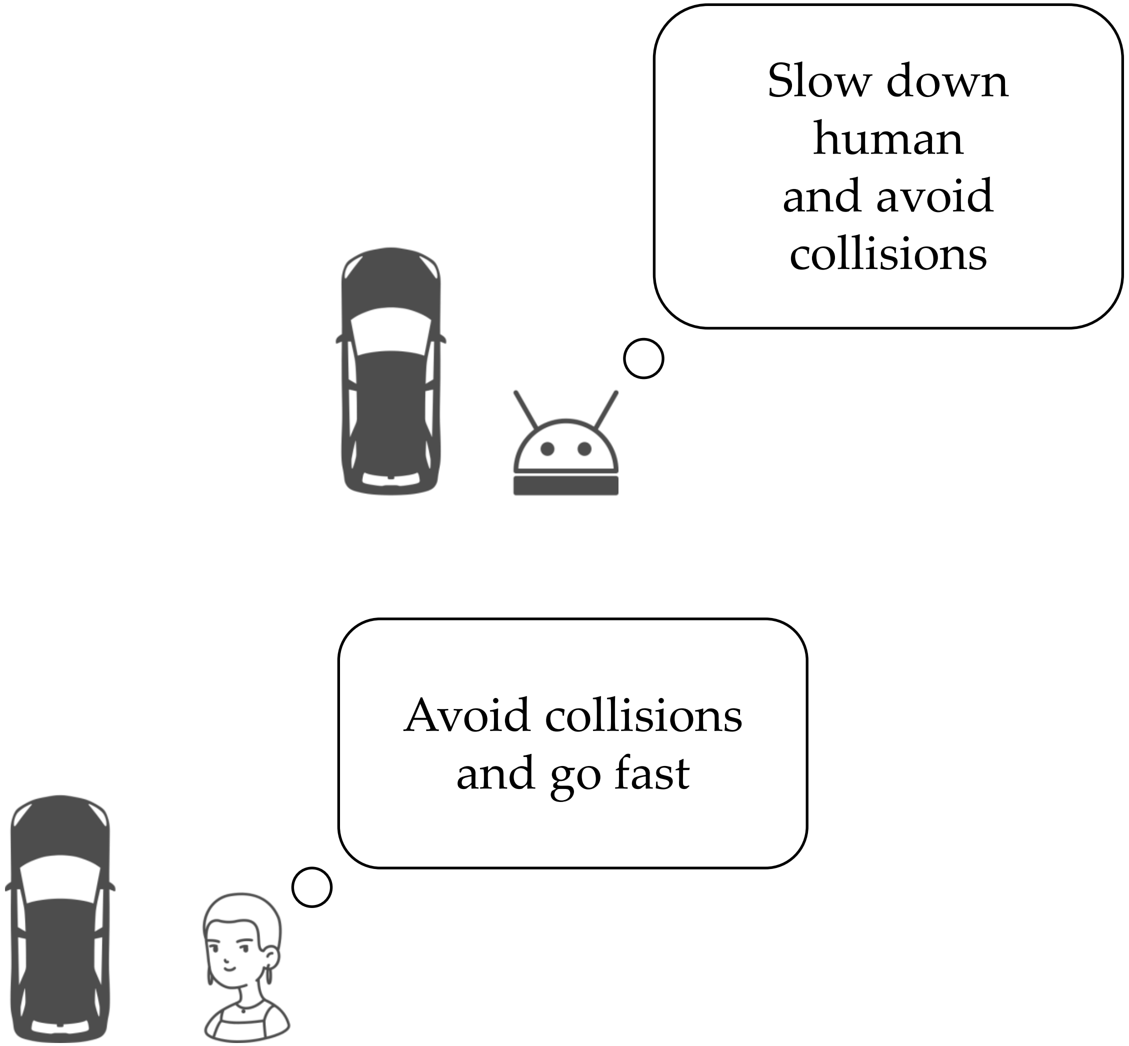
Stackelberg Games

In **Stackelberg game** robot plays first and human chooses best response.

$$a_H^* = \text{argmax}_{\boldsymbol{a_H}} R_H(s^0, \boldsymbol{a_R}, \boldsymbol{a_H})$$

$$a_R^* = \text{argmax}_{\boldsymbol{a_R}} R_R(s^0, \boldsymbol{a_R}, a_H^*(\boldsymbol{a_R}))$$

Robot chooses $\boldsymbol{a_R}$ that leads to best human response (i.e., response where the robot can get the most reward).

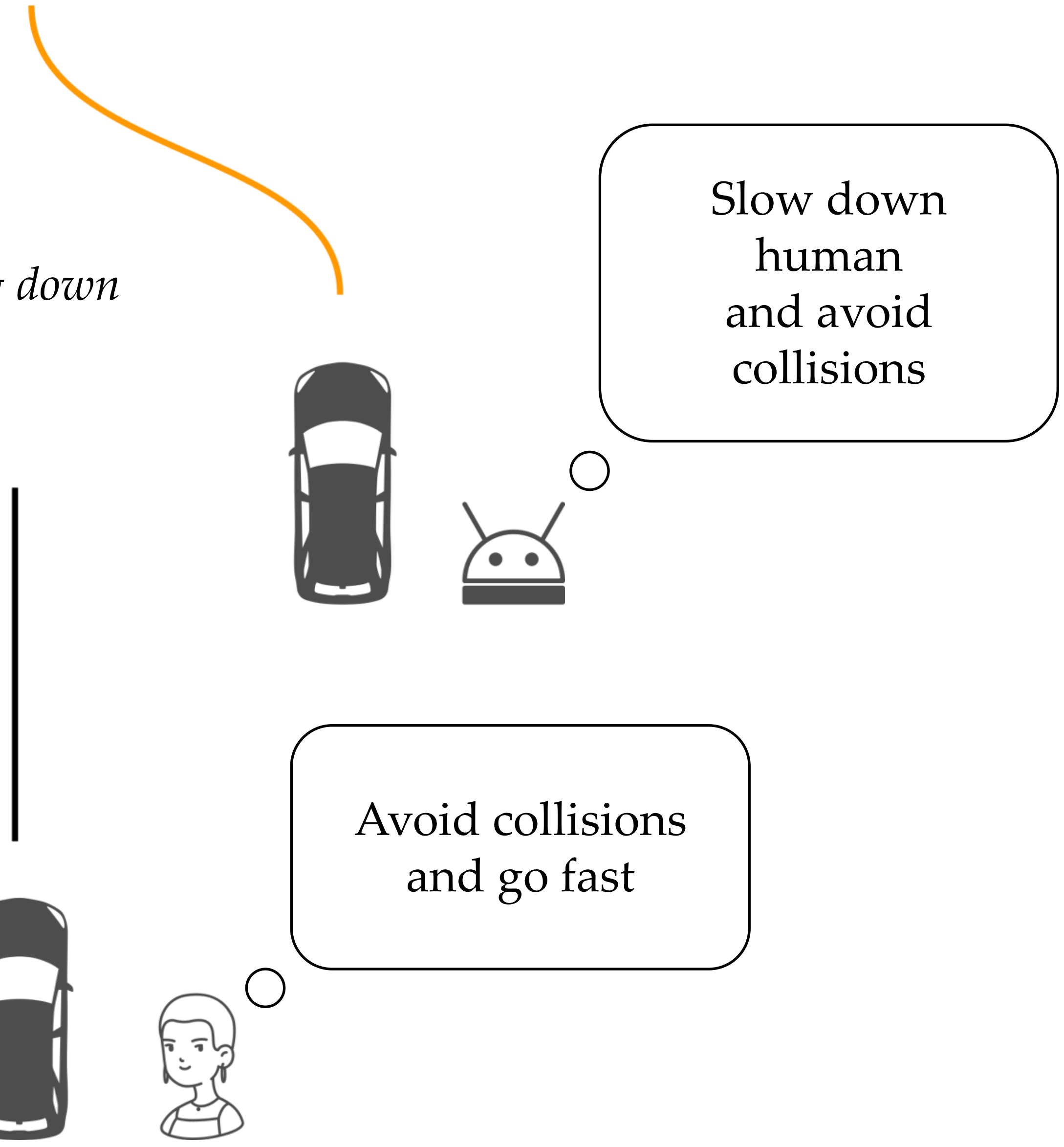

Slow down human and avoid collisions
Avoid collisions and go fast

$$\boldsymbol{a}_R^* = \text{argmax}_{\boldsymbol{a}_R} R_R\left(s^0, \boldsymbol{a}_R, \boldsymbol{a}_H^*(\boldsymbol{a}_R)\right)$$

Block human, their best response to slowing down

$$\boldsymbol{a}_H^* = \text{argmax}_{\boldsymbol{a}_H} R_H\left(s^0, \boldsymbol{a}_R, \boldsymbol{a}_H\right)$$

Slow is best response to blocking




Robots that plan for
human response can
influence humans

Influence

Consider a human-robot setting where the robot gets trajectory reward R

Influence **naturally emerges** as part of the robot's solution when:

1. The robot has a model of human
2. The human changes behavior in response to robot actions

Influence

Consider a human-robot setting where the robot gets trajectory reward R

Influence **naturally emerges** as part of the robot's solution when:

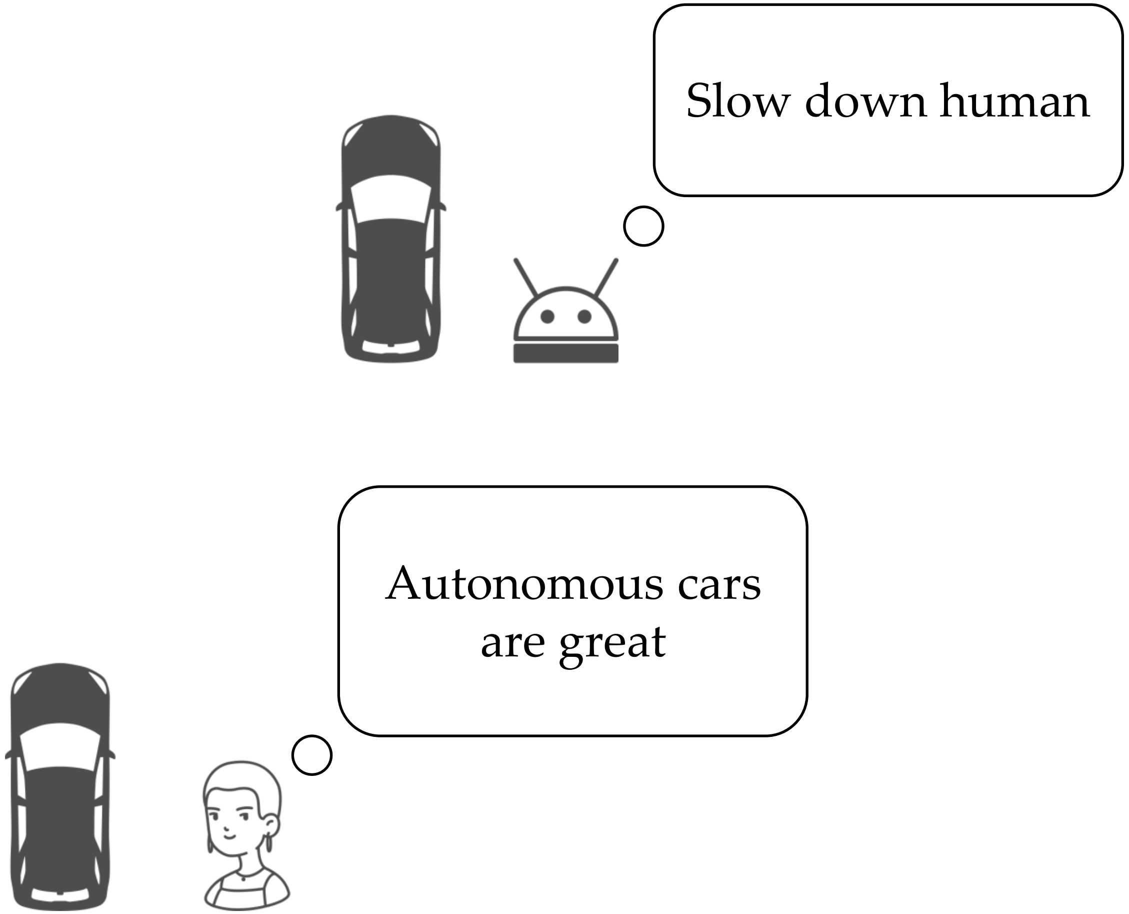
1. The robot has a model of human
2. The human changes behavior in response to robot actions

$$a_H = M\left(s^0, \boldsymbol{a_R}\right)$$

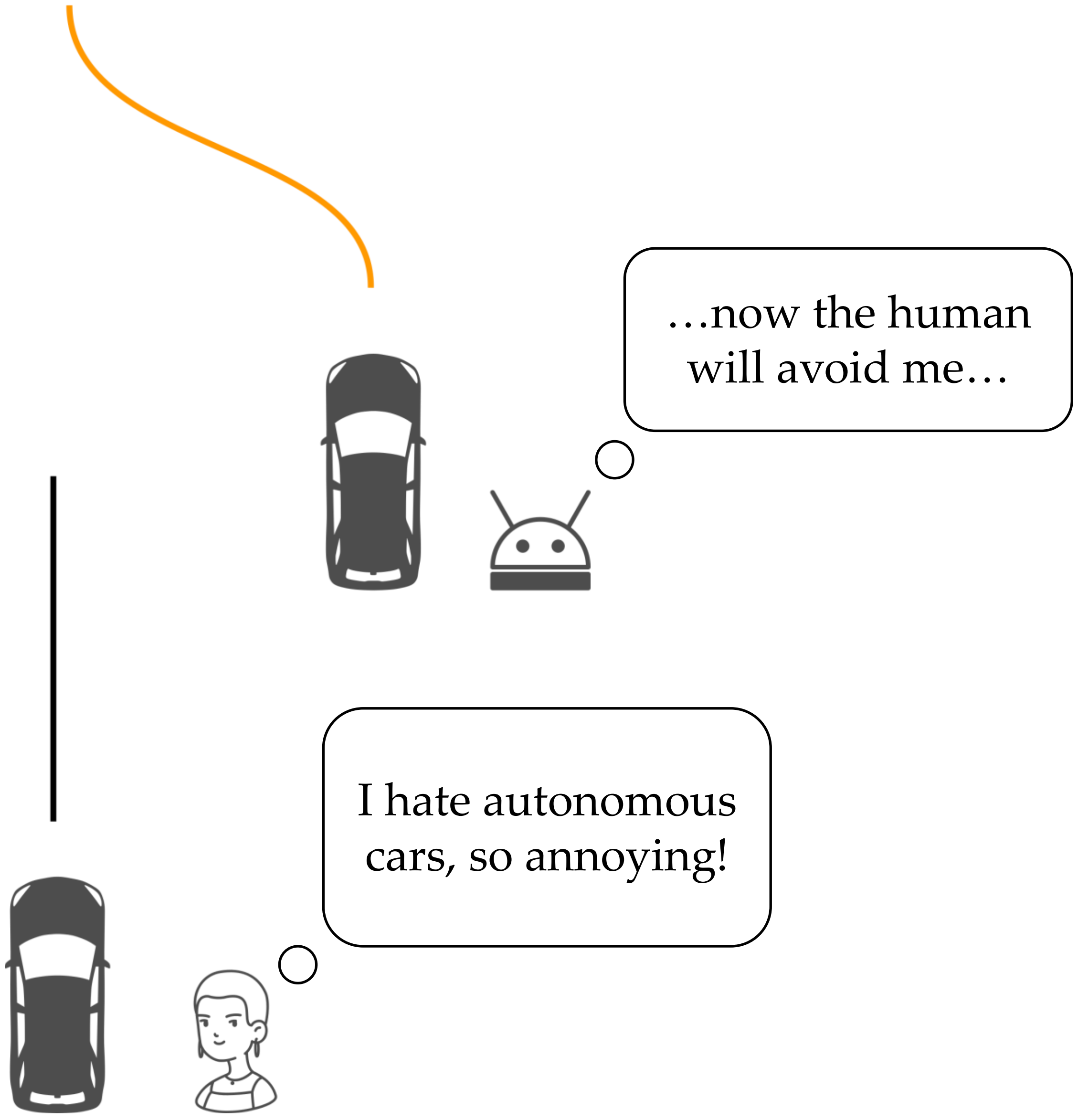
$$a_R^* = \operatorname{argmax}_{\boldsymbol{a_R}} R_R\left(s^0, a_R, M\left(s^0, a_R\right)\right)$$

How does influence affect **trust**?

Influence



Influence


…now the human will avoid me…
I hate autonomous cars, so annoying!

Towards Robots that Influence Humans over Long-Term Interaction

Shahabedin Sagheb, Ye-Ji Mun, Neema Ahmadian, Benjamin A. Christie
Andrea Bajcsy, Katherine Driggs-Campbell, and Dylan P. Losey

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Next Lecture

- Machine teaching + topics you requested